Solutions to Exploration 9.1

1. Is there an association between change in brain volume and type of intervention strategy?
2. The 107 participants in the study
3. Explanatory: Intervention strategy (tai chi, walking, social intervention, control); categorical variable with four categories

   Response: Change in brain volume; quantitative
4. Experiment. Responses are independent between treatment groups.
5. Null hypothesis: The long-run average change in brain volume is the same for each group ($\mu_{\text{Control}} = \mu_{\text{Walking}} = \mu_{\text{TaiChi}} = \mu_{\text{Social}}$)

   Alt hypothesis: The long-run average change in brain volume is different for at least one group (At least one of μ_{Control}, μ_{Walking}, μ_{TaiChi}, μ_{Social} is not the same as the others)
6. a.

Group	Sample size	Mean	SD	Minimum	Lower quartile	Median	Upper quartile	Maximum
Tai chi	29	0.47	0.86	– 1.83	0.01	0.45	0.99	2.20
Walking	27	– 0.15	1.39	– 3.47	– 1.12	– 0.03	0.99	1.83
Social	27	0.41	0.70	– 1.36	– 0.05	0.60	0.81	1.80
No interaction	24	– 0.24	1.26	– 2.04	– 1.18	– 0.59	1.00	2.01

   b. largest increase: Tai chi (average increase of 0.47), largest decrease: control (– 0.24)
7. There does appear to be at least one group different from the others since two interventions saw an increase and two saw a decrease; however the sample sizes are small and the variability within the groups is fairly large.
8. a. Yes, it's possible these differences could occur by chance alone.

   b. 3-S process. Compute a statistic summarizing the evidence. Simulate the null hypothesis of no association through shuffling and recomputing the statistic many (1,000) times. Evaluate strength of evidence by seeing if the observed statistic often or rarely occurs by chance.
9. Does not use all groups.

10. Take absolute value of differences in means between all pairs, then average (MAD statistic for means). Another option is to compare each mean to the overall mean and average them (in absolute value) together.
11. See #10
12. So the differences don't cancel out and it does not matter in which order you subtract.
13. MAD = 0.448
14. Small, positive numbers when null is true; Large, positive numbers when evidence against the null
15. a. 107

    b. Write the changes in brain volume on the cards.

    c. 4 piles

    d. 24, 27, 27, 29

    e. Record the average of the changes in brain volume for each of the 4 groups. From these averages , compute a MAD statistic. Do this many times (1,000 would be good) to get an idea of what the distributions of simulated MAD statistics is assuming the null hypothesis of no association.

    f. Count the number of simulated MAD statistics as extreme as the observed MAD (0.448) and divide by 1,000 to get the p-value.
16. (answers will vary): 0.09, is not more extreme than observed MAD of 0.448
17. (answers will vary): 0.10, is not more extreme than observed MAD of 0.448
18. (answers will vary): 0.220, 0.27, 0.30 – none as extreme as observed (0.448)
19. a. Skewed right b. No, not centered at zero, because the MAD statistic must be positive (can't be negative), so impossible to be centered at zero.
20. a. …equal to 0.448 or larger

    b. 0.04

    c. 0.0313 (after 10,000 simulations)

    d. The p-value is the probability of obtaining a MAD statistic of 0.448 or larger assuming that the null hypothesis (no association between treatment condition and change in brain value) is true.

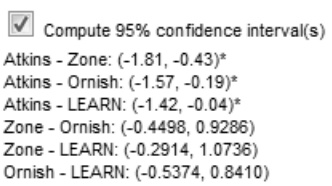
    e. strong
21. a. We have strong evidence of an association between experimental condition and change in brain volume

    b. No, this was an over-all test to see if at least one of the average changes was significant from the others; follow-up analyses are needed to show where differences are.

22. Construct a confidence interval for the difference in average volume changes between the two groups.
23. Can't generalize well because this is not a random sample. Might feel comfortable generalizing to adults in China willing to participate in such a study.
24. Yes, we have evidence of a cause-effect conclusion between type of exercise and change in brain volume because we randomized the subjects to treatments.
25. There may be more accurate ways of determining brain volume that could be explored. It would be nice to get a random sample of elderly Chinese to participate and to have more subjects per group.
26. The study could be replicated with different ethnicities to see if findings are consistent for different ethnic groups.

Solutions to Exploration 9.2

1. Is there evidence of an association between change in BMI and type of diet used?
2. a. Each of the 311 women in the study
   b. Explanatory variable: diet (Atkins, Zone, Ornish, LEARN); categorical with 4 categories
   Response: change in BMI; quantitative
3. a. Random assignment (yes), random sampling (no), Cause and effect is possible but not generalization to a population.
   b. Responses from one treatment group are independent of the responses from another treatment group.
4. Null hypothesis: The long-run mean change in BMI is the same for all three diets
   ($\mu_{\text{Atkins}} = \mu_{\text{Zone}} = \mu_{\text{Ornish}} = \mu_{\text{LEARN}}$)
   Alt: At least one of the means (μ_{Atkins}, μ_{Zone}, μ_{Ornish}, μ_{LEARN}) is different than the others
5. a. Atkins: mean= – 1.65 , SD =2.54 , n =77; Zone: mean = – 0.53 , SD = 2.00 , n = 79 ; Ornish: mean = – 0.77 , SD = 2.14 , n = 76, LEARN: n = 79, Mean = – 0.92, SD = 2.00
   b. The most change in BMI occurs in the Atkins group (– 1.65 average percent change in BMI), while the other three diets appear to have similar change (– 0.92, – 0.77, – 0.53 for LEARN, Ornish and Zone, respectively).
6. MAD = 0.585. p-value = 0.008.
7. We have strong evidence of an association between diet and average change in BMI.
8. a. 3.80, b. p = 0.0106 c. Yes, the p-value is similar and so conclusion is similar.
9. a. Yes. b. Theory-based p-value = 0.0106, the p-values are similar.
10. 18.06/4.75 = 3.80
11. SD = 0.483. The SD of the four group mean weight losses is 0.483.
12. 2.17
13. 77 $0.483^2/2.17^2$ = 3.8
14. Yes, they are all similar. All provide strong evidence of an association between diet and average change in BMI.
15. 95% simultaneous confidence intervals



16. Atkins vs. Zone, Atkins vs. Ornish and Atkins vs. LEARN because they do not include zero.
17. The Atkins diet is associated with significantly more decrease in BMI than the LEARN, Ornish and Zone diets.
18. There is not strong skewness in any of the three groups. The SDs are 2.00, 2.14, 2.00 and 2.54. These SDs are similar – none of them is more than twice the size of any other.
19. Yes, because this is a randomized experiment.
20. Not a random sample, so no obvious population to generalize to. Possibly feel comfortable generalizing to women who are apt to volunteer for this type of a study.
21. Study design looks good, is always nice to get a random sample to be able to generalize more broadly.
22. Expand study to include males. Maybe monitor exercise as well, possibly look at pre/post cholesterol measurements.